## EXPERIMENT 2

**AIM: Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters**

**Objective 1:**

To model coaxial, waveguide or microstrip transmission structures for microwave-frequency operation.

**Objective 2:**

To determine characteristic impedance, propagation constant and transmission characteristics of the selected structure.

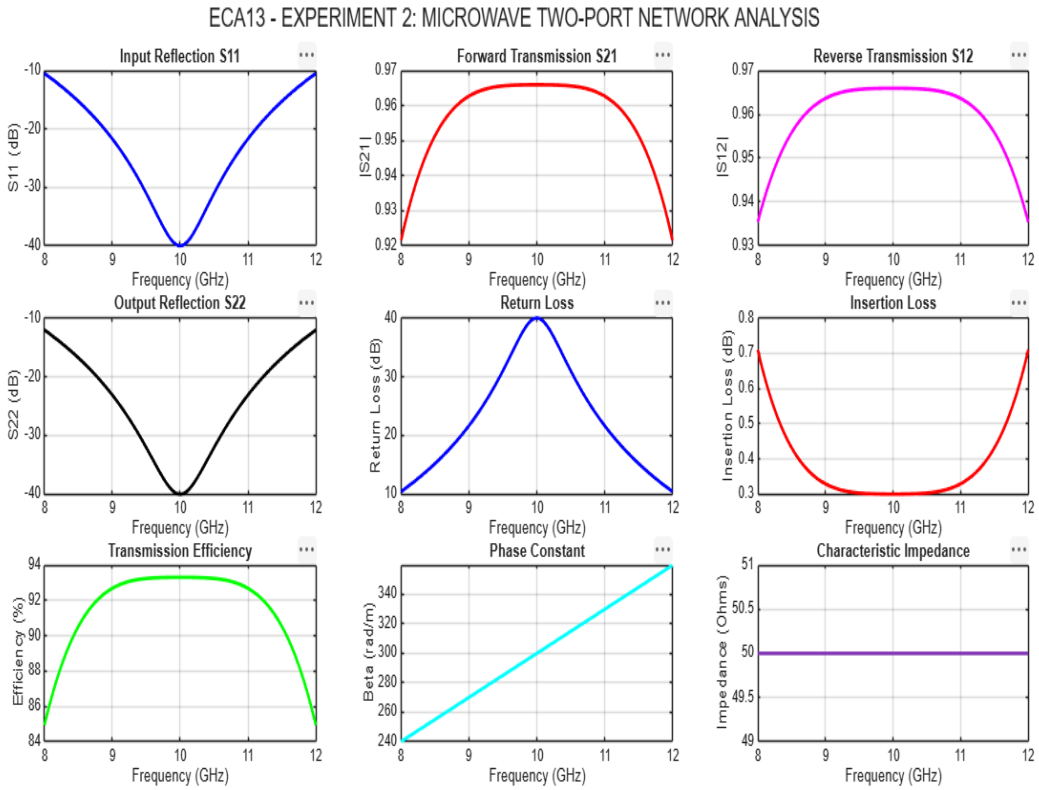
**Objective 3:**

To obtain and analyze $S_{11}$, $S_{21}$, $S_{12}$ and $S_{22}$ over a specified frequency range.

**Objective 4:**

To evaluate return loss, insertion loss and transmission efficiency from the simulated S-parameter response.

**Output:**



## Input parameters

| S.No. | Input Parameter | Symbol | Value |
|---|---|---|---|
| 1 | Speed of Light | $c$ | $3 \times 10^8$ m/s |
| 2 | Characteristic Impedance | $Z_0$ | 50 Ω |
| 3 | Transmission Line Length | $L$ | 50 mm |
| 4 | Center Frequency | $f_0$ | 10 GHz |
| 5 | Starting Frequency | $f_{\min}$ | 8 GHz |
| 6 | Ending Frequency | $f_{\max}$ | 12 GHz |
| 7 | Frequency Step | $\Delta f$ | 10 MHz |
| 8 | Number of Frequency Points | $N$ | 401 |

## Result

The microwave transmission line was simulated over the frequency range of 8–12 GHz. The S-parameters, return loss, insertion loss, transmission efficiency, phase constant, and characteristic impedance were analyzed. The ideal matched, lossless model achieved 100% transmission efficiency with zero reflection.